

ENEL301 – Fundamentals of Engineering Economics and Management

Sustainability

Life Cycle Assessment (2 of 2)

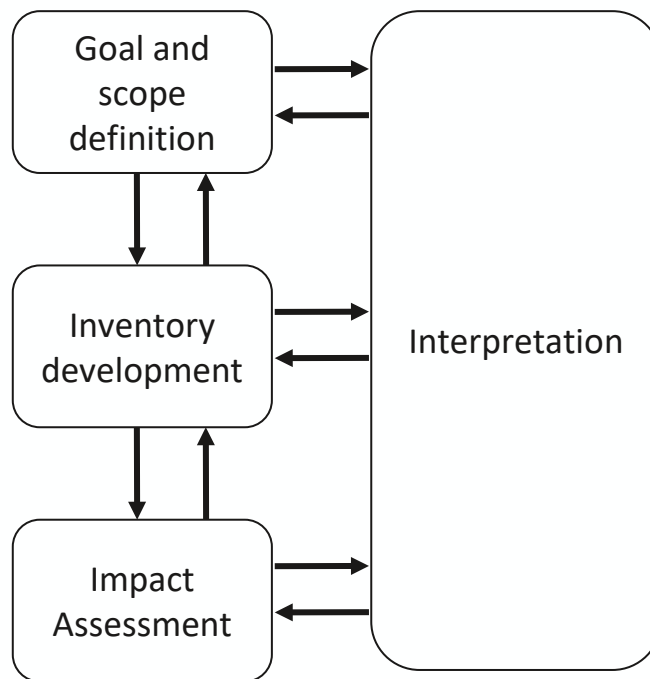
What we are going to cover and when...

Week	Lectures Tuesday 10-11 & Thursday 9-10	Tutorial / Workshop Weeks 1, 4-6 Wednesday 3-4 or 4-5. Weeks 2&3 Wednesday 3-5
1	Introduction to Engineering Practice & Ethics	Ethics
	Collaboration & Teamwork	
2	Māori Perspectives in Engineering	Bicultural Competence and Confidence (timetable and attend one of B01-07 OR B08-14)
	Bias, Diversity & Ethics	
3	Intellectual Property (1/2)	
	Intellectual Property (2/2)	
4	Law in New Zealand (1/2)	IP Protection
	Law in New Zealand (2/2)	
5	Data Privacy and Data Ethics	Law, Data Privacy, Data Ethics
	Business Structures, Strategy, Decisions & Risk	
6	Sustainable Development & Systems Thinking	Sustainable Development & Systems Thinking
	Life Cycle Assessment (1/2)	

What we are going to cover and when...

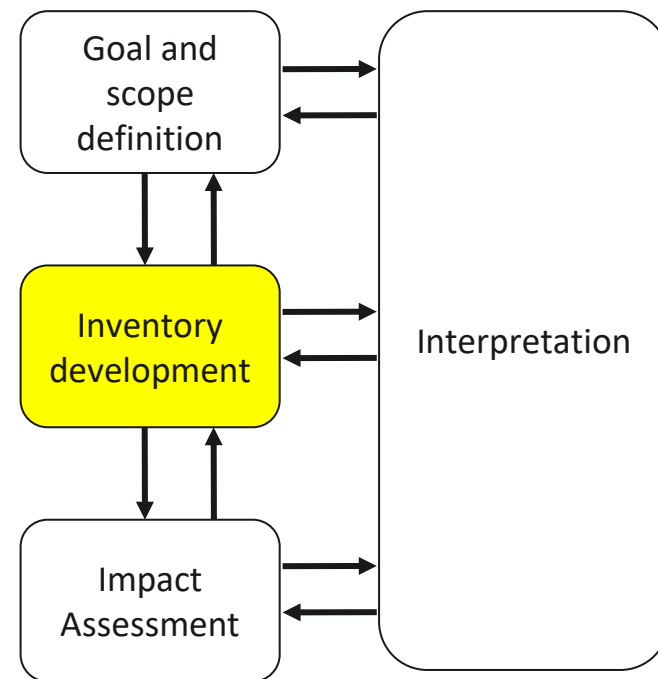
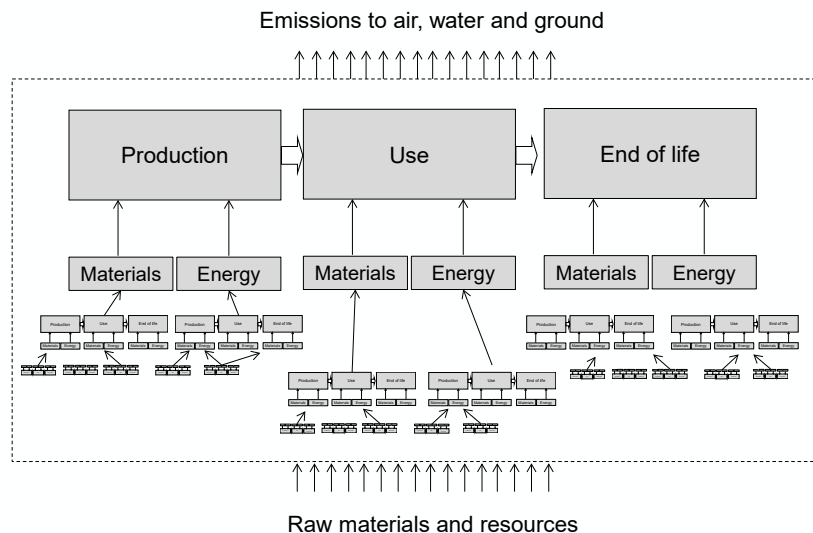
Week	Lectures Tuesday 1-2 & Friday 10-11	Tutorial / Workshop Weeks 7-12 Wednesday 3-4 or 4-5
7	Life Cycle Assessment (2/2)	Life Cycle Assessment
	Social and Economic Assessment	
8	Time Value of Money & Cash Flow (1/2)	Economic Assessment, Time Value of Money & Cash Flow
	Time Value of Money & Cash Flow (2/2) Cost Accounting	
9	Financial Accounting & Reporting (1/2)	Cost Accounting, Financial Accounting, Financial Reporting
	Financial Accounting & Reporting (2/2)	
10	Project Management	Project Management
	Risk Management	
11	Opportunity Analysis	Support for Sustainability Assignment
	Integrated Decisions (1/2) Guest Lecturer – Mike Ullrich (Torqos, formerly Fisher & Pykel Technologies)	
12	Integrated Decisions (2/2)	Risk Management and Opportunity Analysis
	Course Summary	

How to do an LCA



Source: ISO 14040:2006

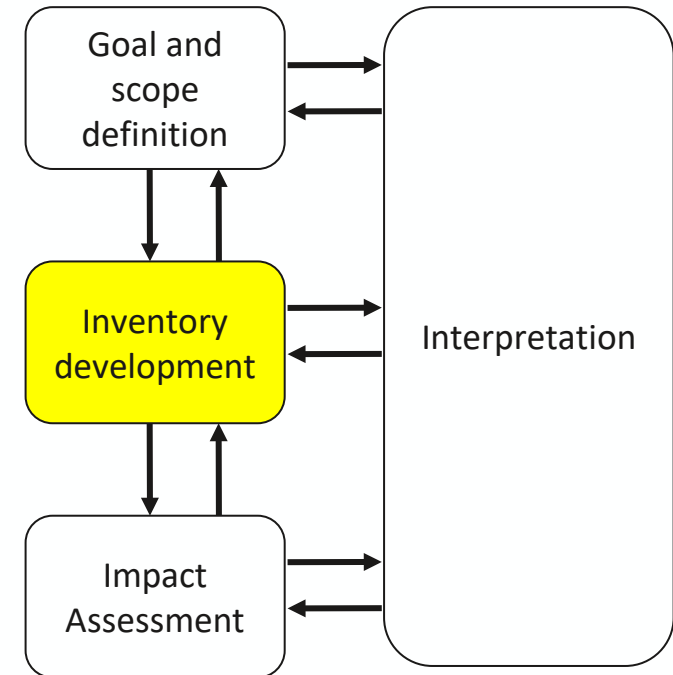
Inventory development



Source: ISO 14040:2006

Inventory development

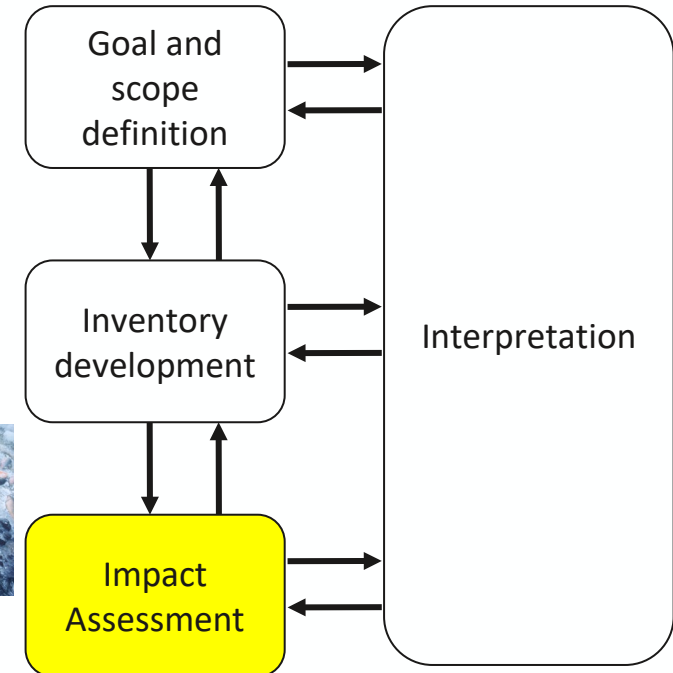
Substance	Compartment	Per functional unit		
		Product A	Product B	Unit
Inflows				
Additives	Raw	2.94	0.43	µg
Air	Raw	406.22	180.04	ng
Aluminum hydroxide	Raw	168.16	180.16	mg
Carbon	Raw	1.28	0.57	mg
Coal washery waste	Raw	0.00	56.95	g
Energy, from solar	Raw	5.76	2.55	kJ
Energy, kinetic, flow, in wind	Raw	0.00	193.63	J
Energy, recovered	Raw	0.00	0.00	J
Energy, unspecified	Raw	369.93	106.59	MJ
Ferromanganese	Raw	505.11	73.32	µg
Granite, in ground	Raw	1.14	0.00	tonne
Graphite, from technosphere, 50MJ/kg	Raw	221.81	32.20	ng
Occupation, industrial area	Raw	0.61	0.29	m2s
Phosphorus pentoxide	Raw	77.64	82.99	mg
Potassium chloride	Raw	12.23	5.42	ng
Secondary glass	Raw	115.43	19.29	µg
Sulfur dioxide, secondary	Raw	321.99	0.25	g
Waste, concrete	Raw	-1.44	-1.66	tonne
Waste, fly ash	Raw	75.00	170.00	kg
bagasse	Raw	29.39	13.57	g
bagasse, 8.7MJ/kg	Raw	572.88	83.16	mg
Energy, from biomass	Raw	4.53	2.01	MJ
etc. etc.				
Outflows				
Carbon dioxide	Air	269.79	230.95	kg
Carbon dioxide, biogenic	Air	256.60	114.33	g
Carbon dioxide, fossil	Air	50.43	22.45	kg
Carbon disulfide	Air	0.80	3.52	mg
Carbon monoxide	Air	379.78	451.32	g
Carbon monoxide, biogenic	Air	0.00	100.41	mg
Carbon monoxide, fossil	Air	0.00	1.42	g
Ethane, 1,1,1-trichloro-, HCFC-140	Air	0.00	0.01	pg
Ethane, 1,1,1,2-tetrafluoro-, HFC-134a	Air	21.37	45.17	mg
Ethane, 1,2-dibromo-	Air	28.27	33.20	µg
Ethane, 1,2-dichloro-	Air	640.86	284.03	pg
1-Methylfluorene	Water	6.71	7.87	µg
2-Hexanone	Water	384.10	451.09	µg
2-Methylnaphthalene	Water	0.93	1.09	mg



Source: ISO 14040:2006

Impact assessment

Substance	Compartment	Per functional unit		
		Product A	Product B	Unit
Inflows				
Additives	Raw	2.94		0.43 µg
Air	Raw	406.22		180.04 ng
Aluminum hydroxide	Raw	168.16		180.16 mg
Carbon	Raw	1.28		0.57 mg
Coal washery waste	Raw	0.00		56.95 g
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Carbon monoxide, biogenic	Air	0.00		100.41 mg
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Ethane, 1,1,1-trichloro-, HCFC-140	Air	0.00		0.01 pg
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Source: ISO 14040:2006

Impact assessment - characterisation

Air emission	Flow kg emitted / functional unit	Characterisation factor kg CO ₂ -eq/ kg emitted	Characterised result kg CO ₂ -eq / functional unit
CO ₂	0.1	1	0.1 x 1
CH ₄	0.09	21	0.09 x 21
N ₂ O	0.00012	310	0.00012 x 310
Sum			2.0272

Impact assessment - characterisation

Inventory

Compartment	Elementary flow	Amount (kg/ functional unit)
Raw	Air	0.01563
Raw	Aluminium, 24% in bauxite, 11% in crude ore, in ground	2.11E-02
Raw	Aluminum hydroxide	2.11E-02
Air	Cyclohexane	3.91E-08
Air	Diethylamine	1.47E-20
Air	Dimethyl malonate	7.86E-22
Air	Dinitrogen monoxide	2.35E-06
etc.		

X

Indicator 1

Compartment	Elementary flow	Characterisation factor (\$/kg)
Raw	Coal, 13.3 MJ per kg, in ground	0.01563
Raw	Coal, 18 MJ per kg, in ground	2.11E-02
Raw	Coal, 18.0 MJ per kg, in ground	2.11E-02
etc.		

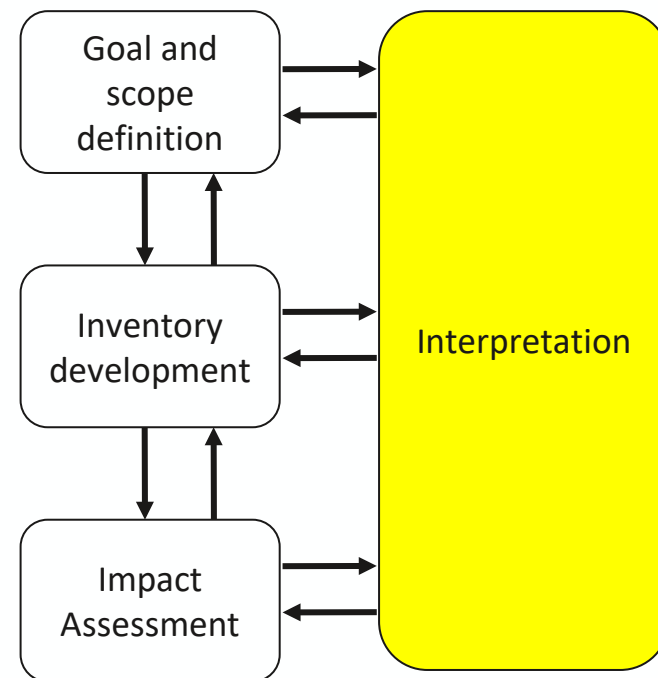
Indicator 2

Compartment	Elementary flow	Characterisation factor (kg NMVOC/kg)
Air	1-Butanol	1.05
Air	1-Butene	1.82
Air	1-Butene, 2-methyl-	1.3
etc.		

X

Interpretation

- "Which is better?" can depend
- Identify the reasons for your results → answer "Why?" 5 times
- Check methods inc. assumptions, data quality
- Sensitivity studies (Full LCA studies); effect of key assumptions on your results
- Uncertainty studies (Full LCA studies); Monte Carlo simulations to evaluate effect of data uncertainty on results
- Communicate your limitations, uncertainty, trade-offs

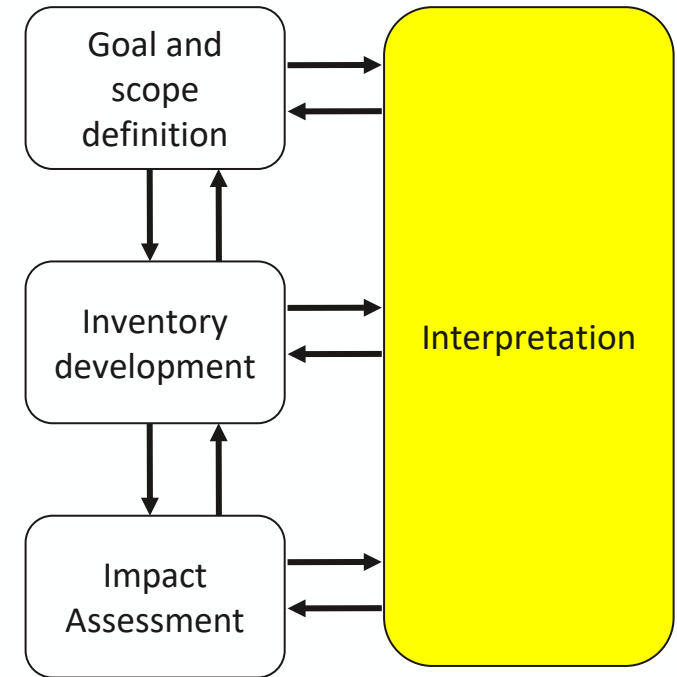


Source: ISO 14040:2006

Interpretation



- Burden shifting occurs when a solution reduces one environmental impact but increases another.



Source: ISO 14040:2006

Practicalities of LCA

Factor	Streamlined LCA Basic	Streamlined LCA with tool-support	Full LCA
Typical use	Get a feel for the issues	Quick decisions, single product or comparisons	Detailed decisions or comparative claims
Standard compliance	None	Typically aligned to ISO 14040/14044	ISO 14040 / 14044 (or other international standard)
Data	Generic, publicly available	Generic, limited dataset	Supply-chain specific
Software	Spreadsheets	Online or integrated tools (e.g. SolidWorks)	OpenLCA, Sphera (GaBi), SimaPro
Resolution	Low	Medium	High to very high
Uncertainty	Very high	Medium to High	Low
Expertise level needed	Low	Medium to high	High to very high
Time taken	Hours	Days to weeks	Months to years
Cost (for software + expertise)	Just your time	\$100s → \$1000s	~\$15 k → \$100 k



Your assignment based at this level

Classification: In-Confidence

Practicalities of LCA

Streamlined LCA - basic

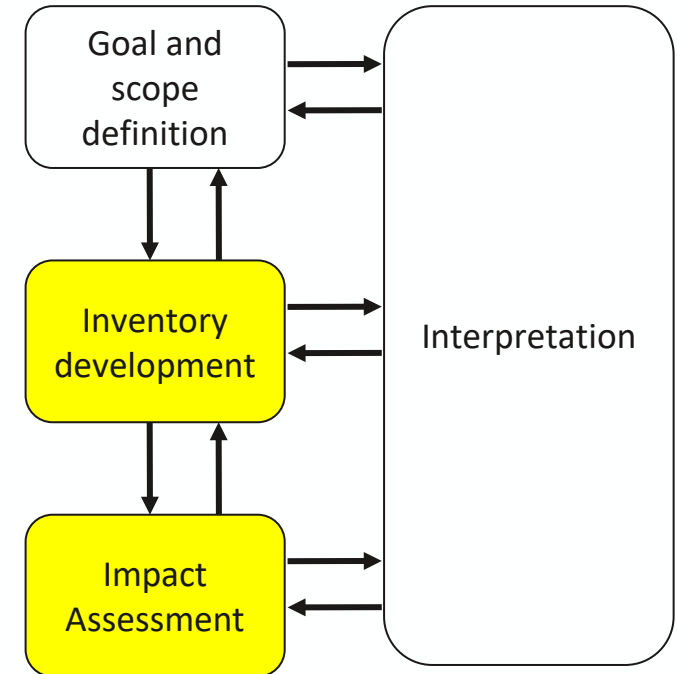
$$E = Q \times F$$

E = Emissions from the emissions source (e.g. kg CO₂-eq)

Q = Activity data, e.g. quantity of material used (e.g. kg)

F = Emission factor for emissions source (e.g. kg CO₂-eq / kg)

1. Set functional unit, system boundary
2. Collect data
3. Use $E = Q \times F$ in a spreadsheet, add everything up
4. Produce a report



Source: ISO 14040:2006

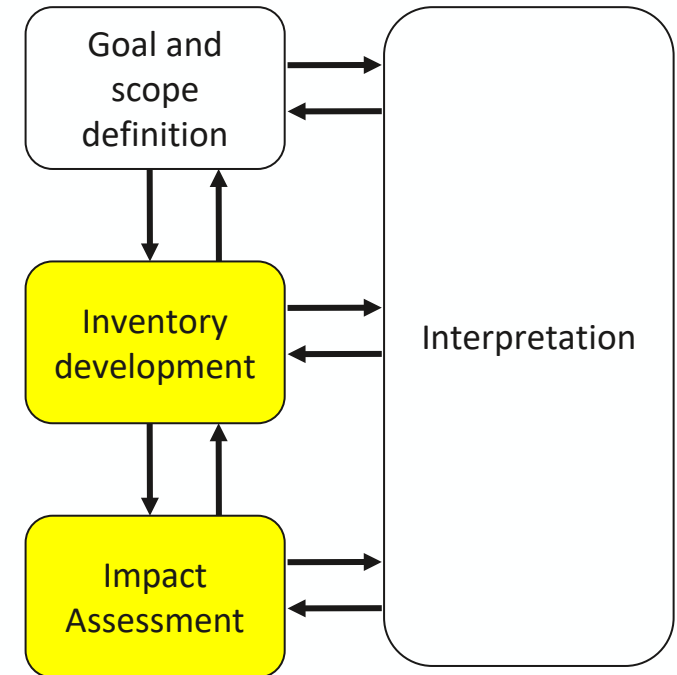
Practicalities of LCA

Streamlined LCA - basic

- Emission factors for electricity / fuel use / transport etc.
- [Measuring emissions catalogue](#)

Table 5.2: Emission factors for purchased grid-average electricity – annual average

Emissions Source	Unit	kg CO ₂ -e/unit	CO ₂ /unit (kg CO ₂ -e)	CH ₄ /unit (kg CO ₂ -e)	N ₂ O/unit (kg CO ₂ -e)
Annual Averages					
Electricity Used – 2025	kWh	0.0786625	0.0761325	0.00240719	0.000122795



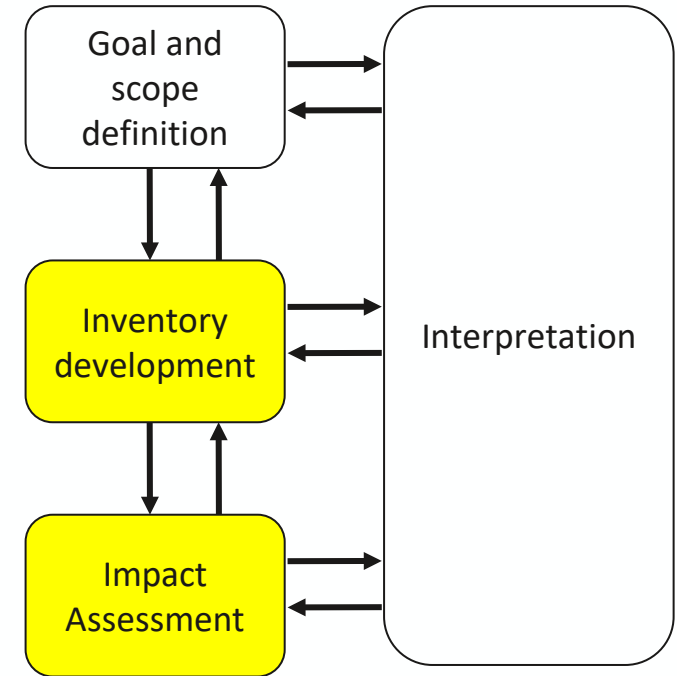
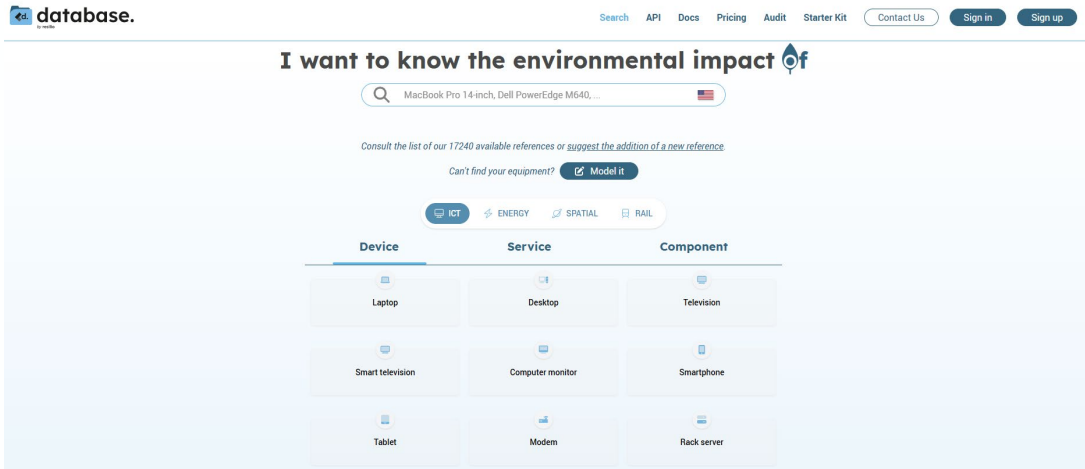
Source: ISO 14040:2006

Practicalities of LCA

Streamlined LCA - basic

resilio – new database for electronics

- <https://db.resilio.tech/>
- Emission factors referenced in your assignment



Source: ISO 14040:2006

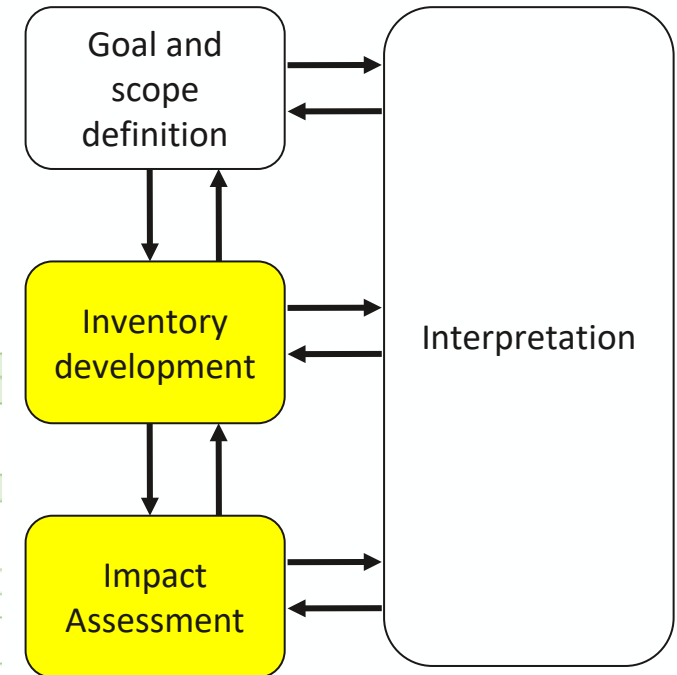
Practicalities of LCA

Streamlined LCA - basic

The Inventory of Carbon and Energy (ICE) Database

- <https://circularecology.com/embodied-carbon-footprint-database.html>
- Emission factors for material production
- Shaping processes may or may not be included

Materials	Embodied Carbon Kg CO ₂ /sqm
PV Modules	
Monocrystalline	242 (132 to 440) CO ₂ Only
Polycrystalline	208 (99 to 289) CO ₂ Only
Thin Film	67 (40 to 92) CO ₂ Only
Roads	
Asphalt road - Hot construction method - 40 yrs	99 KgCO₂/Sqm
Construction	32.8 KgCO ₂ /Sqm
Maintenance - 40 yrs	12.3 KgCO ₂ /Sqm
Operation - 40 yrs	54.0 KgCO ₂ /Sqm
Asphalt road - Cold construction method - 40 yrs	97 KgCO₂/Sqm
Construction	28.2 KgCO ₂ /Sqm
Maintenance - 40 yrs	14.8 KgCO ₂ /Sqm
Operation - 40 yrs	54.0 KgCO ₂ /Sqm



Source: ISO 14040:2006

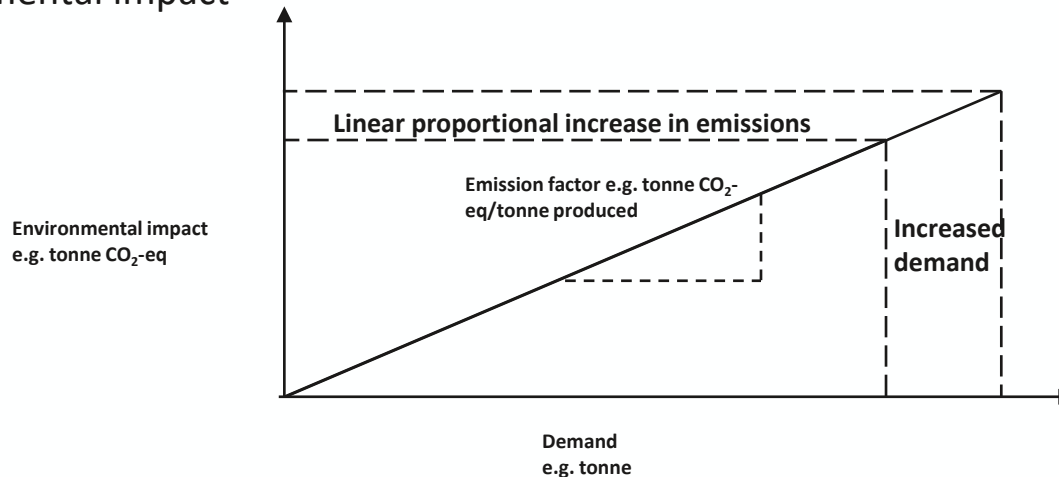
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Time taken	Hours	Days to weeks	Months to years
Cost (for software + expertise)	Just your time	\$100s → \$1000s	~\$15 k → \$100 k

Some nuances of doing LCA

Attributional vs consequential

- Attributional LCA assumes that increased demand will result in the same (historical) environmental impact



Some nuances of doing LCA

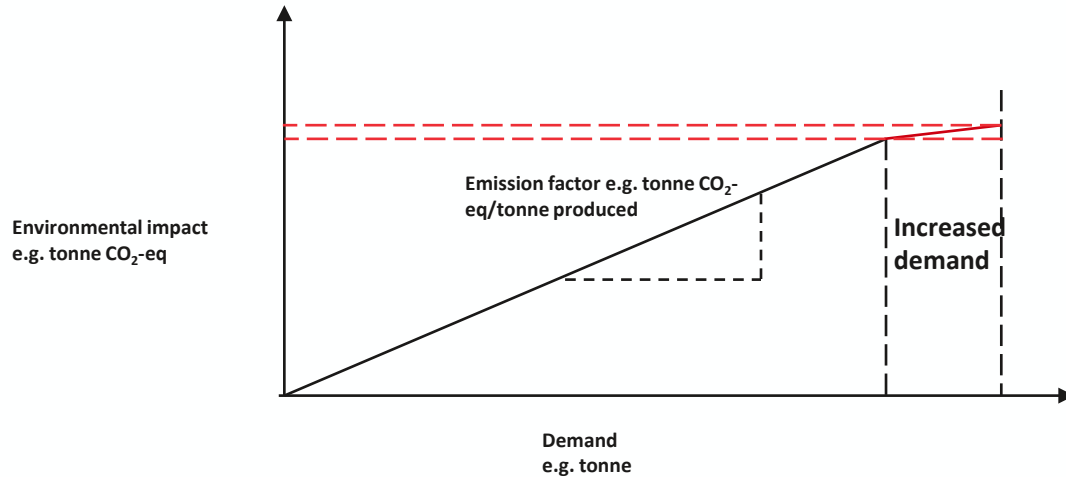
Attributional vs consequential

- Consequential LCA accounts for consequences arising from demand change
 - Displacement effects in the economy
- Supply & demand
 - In a supply-constrained market, demand increases prices at the “supply margins”
 - Higher prices with constrained supply → someone will be choose a cheaper alternative
 - The cheaper alternative can have a different environmental impact to the average
 - Can cause ‘ripple’ of consequential effects, and unintended outcomes

Some nuances of doing LCA

Attributional vs consequential

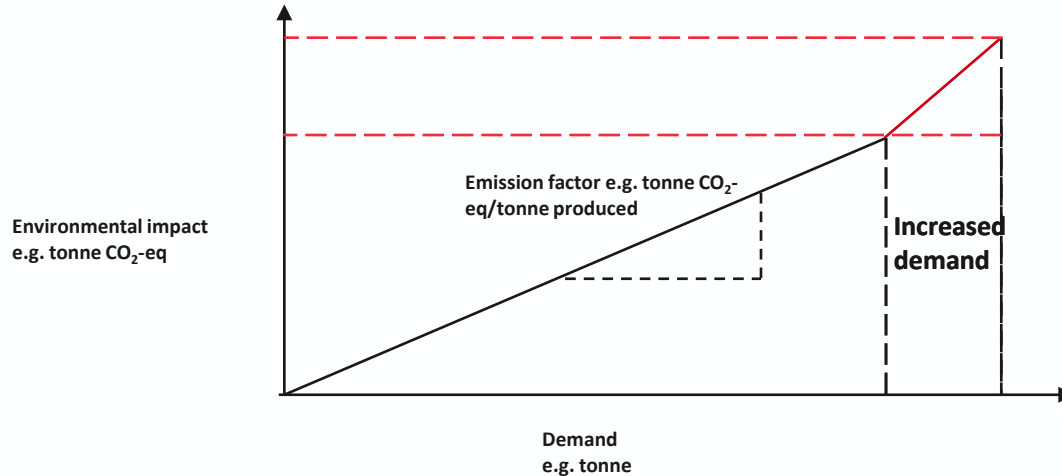
- Consequential LCA accounts for consequences arising from demand change
- When marginal supply is better than average



Some nuances of doing LCA

Attributional vs consequential

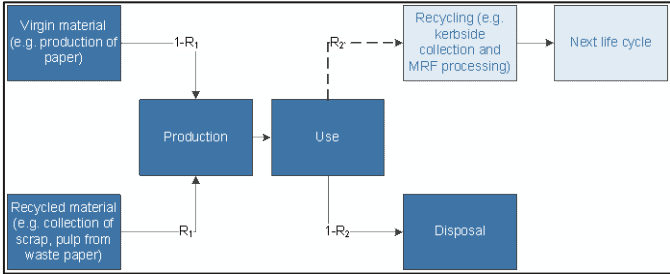
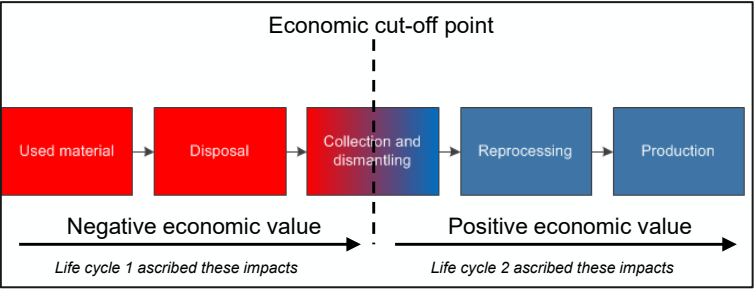
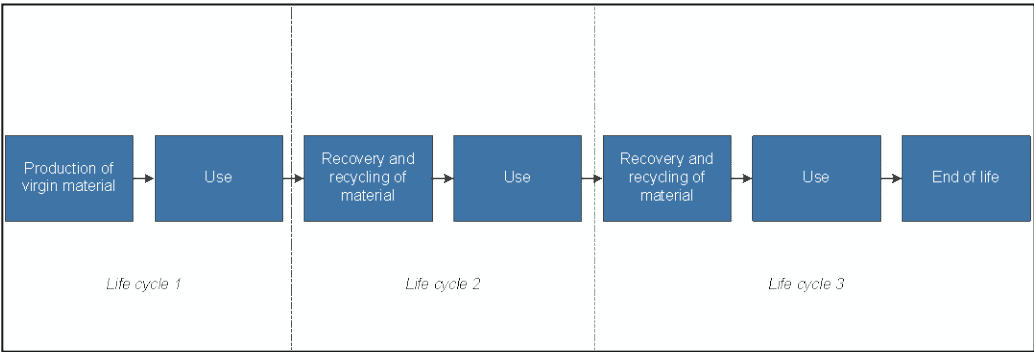
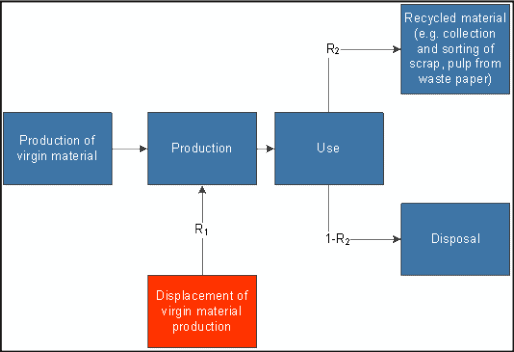
- Consequential LCA accounts for consequences arising from demand change
- When marginal supply is worse than average



Environmental outcomes cannot be decoupled from the economy

Some nuances of doing LCA

Approaches to recycling



Some nuances of doing LCA

Input-Output LCA; 'consumption' modelling

- Economic sectors have their own environmental footprint
- Systems of National Accounts model monetary exchanges within the economy
- Use \$ value to model environmental impact
- Like the $E = Q \times F$ approach, but F = emissions per dollar spent
- System boundary is the economy; captures 'everything'
- But....
 - Not auditable / traceable
 - Limited to boundary of the economy (e.g. NZ)
 - Resolution can be poor
 - Price inflation or discounting does not necessarily directly translate into increased/decreased impact

Some nuances of doing LCA

Input-Output LCA; ‘consumption’ modelling

Consumption Emission Intensities			
Supplementary material for the report: Market Economics Limited, 2023, Consumption Emissions Modelling, report prepared for Auckland Council. The information provided in this document are to be read with the assumptions and limits set in the report.			
Code	Household Economic Survey Class Description	t CO2-e per \$	kt CO2-e per \$m
01.1.01.1	Citrus fruit (fresh or chilled)	4.16E-04	0.4160
01.1.01.2	Bananas (fresh or chilled)	4.16E-04	0.4160
01.1.01.3	Apples and pears (fresh or chilled)	3.65E-04	0.3650
01.1.01.4	Stone fruit (fresh or chilled)	4.16E-04	0.4160
01.1.01.5	Berry fruit, kiwifruit and grapes (fresh or chilled)	4.20E-04	0.4200
01.1.01.6	Other fresh or chilled fruit	4.16E-04	0.4160
11.3.02.2	Other personal effects	3.76E-04	0.3760
11.4.01.0	Life insurance	1.72E-05	0.0170
11.4.02.0	Dwelling insurance	1.20E-05	0.0120
11.4.03.0	Contents insurance	1.20E-05	0.0120

<https://www.knowledgeauckland.org.nz/publications/consumption-emissions-modelling/>

- Hybrid LCA combines traditional ‘process’ modelling with IO-LCA
 - Useful for filling in gaps, e.g. services

Homework

- Reading:

Finnveden, G., Hauschild, M. Z., Ekvall, T., Guinée, J., Heijungs, R., Hellweg, S., ... & Suh, S. (2009). Recent developments in life cycle assessment. *Journal of Environmental Management*, 91(1), 1-21.
<https://doi.org/10.1016/j.jenvman.2009.06.018>

In particular, read and take notes on these sections:

1. Introduction
2. Some characteristics of Life Cycle Assessment vis-a`-vis other environmental assessment tools
3. Attributional and consequential LCA
- 6.3 Input–output and hybrid LCA
10. Discussion

Take-aways

1. Goal and scope definition
 2. Inventory development
 3. Impact Assessment
 4. Interpretation
- Getting the basics right is critical; goal and scope definition, and inventory data
 - Streamlined LCA uses $E = Q \times F$, to minimise complexity and time, but at the expense of certainty, resolution
 - Careful interpretation needed, particularly when you overlay economic considerations

Reminders and where you should be at...

Sustainability Assignment. By the end of this week, you should have:

- Met your team, worked out your ways of working, worked out a plan
- Read and interpreted the brief
- Completed some modelling and writing; causal loop diagram
- Started other modelling: life cycle assessment, economic impacts
- First round of peer-assessment next week
- **Complete the survey on team behaviours (under Assessment → Sustainability Assignment)**

Test tonight; C1/C2 (check your timetable). Be there for a 6:30 pm start

LCA workshop this Wednesday; bring your laptop / tablet (you'll be using Excel)